MYFROYOLAND CASE: OPERATIONS ANALYSIS

Operations Management Mid-Term Assignment

Q1: Process Flow Diagram, Capacity, and Bottleneck

Process Flow Diagram

ENTER → Step 1: Choose Cup (10 sec) → Step 2: Dispense Yogurt (80 sec, 5 machines) → Step 3: Toppings Bar (60 sec, 4 positions) → Step 4: Weigh & Pay (50 sec, 1 counter) → EXIT

Capacity Calculation for Each Stage

Step	Activity Time	Resources	Capacity/hr
1. Choose Cup	10 sec	Ample	360 (3600sec/10sec) *1
2. Dispense Yogurt	80 sec	5 machines	225
3. Toppings Bar	60 sec	4 positions	240
4. Weigh & Pay (BOTTLENECK)	50 sec	1 counter	72 ← LIMIT

Overall Process Capacity: 72 customers per hour

The weighing counter is the only step with just one resource and one staff member. It limits the entire store to 72 customers/hour. Everything else has excess capacity.

Q2: Weekend Peak Demand, Turnaway, and Lost Revenue

Can the Store Serve Peak Demand?

Peak Arrival Rate	90 customers/hour
Store Capacity	72 customers/hour
Verdict	NO — Cannot handle all demand

Customers Turned Away and Lost Revenue

Turnaways per hour: 90 - 72 = 18 customers

3-hour peak evening: 18 × 3 = 54 customers turned away

Average customer spend: ₹225

Lost revenue per evening: 54 × ₹225 = ₹12,150

Peak weekends per month: 12

Monthly turnaways: 54 × 12 = 648 customers

Monthly lost revenue: 648 × ₹225 = ₹145,800

The store is leaving approx ₹146,000 on the table every month because the weigh-and-pay counter can't keep up.

Q3: Theoretical Flow Time and Little's Law

Theoretical Flow Time (No Waiting)

Step	Activity	Time (sec)
1	Choose cup	10
Walk	Between stages	8
2	Dispense yogurt	80
Walk	Between stages	8
3	Toppings bar	60
Walk	Between stages	8
4	Weigh & pay	50
Walk	Exit	6
TOTAL		230 seconds

Theoretical flow time: 3 minutes 50 seconds (with no waiting at any step)

Little's Law Analysis at 60 Customers/Hour

Little's Law: $L = \lambda \times W$ ----------check

L = average customers in store, λ = arrival rate, W = time in system

NO queuing (ideal): $L = 60 \times (3.83/60) = 3.83$ customers in store

This means if there were no queues, only about 4 people would be in the store at any time. But on a Saturday night, Ananya saw above 20 people. On Saturday nights, 90 customers arrive per hour but only 72 can be served. That means 18 customers per hour queue up waiting for the counter. Most people aren't moving through — they're stuck waiting. Using Little's Law again: if average wait is 10 minutes at the counter, then total time in system is 3.83 + 10 = 13.83 minutes. So $L = 90 \times (13.83/60) = 20.7$ customers in store. This matches what Ananya saw — chaos, not flow.

Q4: M/M/1 Queue Model at Varying Arrival Rates

Queue Model Setup: 1 server (counter), service rate = 72 customers/hour

Calculations at λ = 60 customers/hour

Metric	Result
Utilization (ρ)	0.833 (83.3%)
Wait in queue (Wq)	4.17 minutes
Customers in queue (Lq)	4.17 customers

about 4 people waiting.

Calculations at λ = 65 customers/hour

Metric	Result
Utilization (ρ)	0.903 (90.3%)
Wait in queue (Wq)	7.75 minutes
Customers in queue (Lq)	8.38 customers

Approx. 8 customers

Calculations at λ = 70 customers/hour

Metric	Result
Utilization (ρ)	0.972 (97.2%)
Wait in queue (Wq)	29.2 minutes
Customers in queue (Lq)	34.03 customers

At 97% utilization, customers wait nearly half an hour. The queue explodes.

The Cliff Effect — What Happens as Utilization Approaches 100%

Utilization	Arrival Rate	Wait Time	Queue Size
83.3%	60/hr	4.17 min	4.17
90.3%	65/hr	7.75 min	8.38
97.2%	70/hr	29.2 min	34.03
99%	71/hr	~116 min	~136

Waits increase exponentially as utilization approaches 100%. Running a single counter near maximum capacity doesn't work, it would create catastrophic slowdowns.

Q5: Adding a Second Weighing Counter (M/M/2 System)

With a second counter and staff member during weekend peaks:

Metric	Current (1 Counter)	With 2nd Counter
Capacity	72/hr	144/hr
At 90/hr peak: Wait time	Cannot serve	**32 seconds**
Customers in queue	**>30**	**~0.8**
Server utilization	**125% (overflow)**	**62.5% (healthy)**

Revenue Impact

Extra customers served at peak: 18/hour × 3-hour evening × 12 weekends = 648/month

Revenue per customer: ₹225

Additional monthly revenue: 648 × ₹225 = ₹145,800

Cost of second staff member: ~₹12,000/month

Return on investment: ₹145,800 ÷ ₹12,000 = 12.2× payback within first month

Under What Conditions Does Adding a Second Staff Member Pay for Itself?

The payback depends on three factors: (1) the hourly wage of the second staff member, (2) whether weekend peak demand stays at 90/hr or higher, and (3) the profit margin on each sale. At ₹225 per customer and ₹12,000/month wage, the second counter pays for itself in the first peak weekend.

Final Recommendation for Ananya

Add a second weighing counter during weekend peak. Every evening the store operates at 90/hr arrivals with only one counter, it losses ₹12,150. Every month, it's ₹145,800. The second counter costs a fraction of that and creates a fundamentally different experience for customers. Bottleneck is an issue to solve.